UE24CS352A Machine Learning Mini-Project

SongNet Reproduction: Real-Time Music Genre Classification

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Problem statement

Streaming platforms need to tag the genre of new music automatically. SongNet (Zhang, Zhang and Chen, Stanford) proposes a convolutional recurrent neural network (C-RNN) that classifies genre directly from mel-spectrograms and outputs a genre prediction at every time step, so a song can be classified while it plays. The paper reports 65.23% test accuracy on FMA-small, against 46.38% for the best of five traditional baselines trained on pre-computed features. Our goal was to reproduce all six models under one shared, leakage-free protocol and test whether the C-RNN's advantage holds.

Dataset

- **FMA-small** from the Free Music Archive: 8,000 MP3 clips of 30 seconds, 8 balanced genres with 1,000 clips each (Electronic, Experimental, Folk, Hip-Hop, Instrumental, International, Pop, Rock). Labels come from the genre_top column of FMA's tracks.csv.
- **Cleaning:** 6 MP3s that are empty or truncated were removed, leaving 7,994 tracks.
- **Split:** stratified 70% / 20% / 10% with seed 42, giving 5,595 training, 1,599 validation and 800 test tracks (100 per genre). The split is saved as splits.csv and loaded by every notebook; each notebook prints the file's checksum to confirm all models used identical tracks.
- **Baseline input:** 140 MFCC statistics from FMA's features.csv (20 coefficients x 7 statistics). The paper's 140 features also include metadata such as release year and number of listens, but it does not list the columns, so we used audio features only. No model in our reproduction uses metadata.
- **C-RNN input:** a 128-band log-mel spectrogram per clip (22,050 Hz, FFT size 2,048, hop 512), giving a 128 x 1,292 array.

Approach

Baselines. Random Guess, k-nearest neighbours, logistic regression, multilayer perceptron and linear SVM in scikit-learn. Features are standardised using training statistics only. Hyperparameters are chosen on the validation split, whereas the paper used scikit-learn defaults.
C-RNN. Three convolutional blocks (convolution, batch normalisation, ReLU, max-pooling, dropout 0.3) turn the spectrogram into a sequence of 161 time steps. A 2-layer GRU with 128

hidden units reads the sequence, a linear layer gives 8 genre scores at every step, and the scores are averaged over time to give the song-level prediction. Training uses Adam with learning rate 0.001 and ReduceLROnPlateau, as in the paper, with batch size 32 and early stopping on validation accuracy.

Differences from the paper. The paper applies 1-D convolutions along the time axis and, after trying an LSTM, uses a time-distributed dense softmax layer for its sequence stage. We used 2-D convolutions and a GRU, and we average logits rather than softmax probabilities. Filter counts, kernel and pooling sizes, and the spectrogram settings are not given in the paper and are our choices.

Implementation

The project consists of three standalone Google Colab notebooks: random_knn_logreg, mlp_svm and CRNN. Each downloads FMA, loads the shared split, trains its models, evaluates them once on the test split with the same metrics (accuracy, macro precision, recall and F1, per-genre report, confusion matrix) and saves results to Google Drive.

The C-RNN is implemented in PyTorch and trained on a T4 GPU with mixed precision. Spectrograms are cached to disk once as float16. The model has about 440,000 parameters; an epoch takes about 24 seconds. Validation accuracy peaked at epoch 16 and training stopped at epoch 24.

Results

Test accuracy on the same 800 test tracks for every model:

Model	Paper	Ours	Our macro F1
Random Guess	12.50%	12.75%	0.128
K-Nearest Neighbors	36.38%	51.75%	0.500
Logistic Regression	42.25%	54.12%	0.528
Multilayer Perceptron	44.88%	56.75%	0.560
Linear SVM	46.38%	53.50%	0.508
C-RNN	**65.23%**	**63.25%**	**0.626**

Conclusions

- **The C-RNN result reproduces.** 63.25% against the paper's 65.23%; with 800 test tracks the 95% margin is about 3.4 points, so the 2-point gap is within sampling error.
- **The baselines are stronger than reported.** Every baseline beat its paper number by 7 to 15 points while using audio features only, most likely because we tuned and standardised them instead of using defaults. As a result the C-RNN leads the best baseline (MLP) by 6.5 points, an 11.5% relative gain, not the paper's 41%.
- **Learned features still win.** A model that reads the spectrogram directly beats every model built on hand-crafted features.
- **Pop and Experimental are the hardest genres**, in the paper and in our results (C-RNN recall 27% and 50%). Both borrow characteristics from other genres; Experimental songs were most often predicted as Instrumental.
- **Limitations and next steps:** results come from a single seed; our C-RNN differs from the paper's 1-D convolution and time-distributed design; real-time latency was not measured.

Next steps are a variant matching the paper's architecture, runs over several seeds, data augmentation, and measuring accuracy from partial clips to test the real-time claim.

References

Zhang, C., Zhang, Y., and Chen, C. SongNet: Real-time Music Classification. Stanford University.
Defferrard, M., Benzi, K., Vandergheynst, P., and Bresson, X. FMA: A Dataset for Music Analysis. ISMIR 2017.